SMARTBRIDGE SKILLWALLET

Coding & Solution

Date	25 September 2026
Team ID	SWTID-2026-4307
Project Name	AI Weather Wise API
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/SWTID-2026-4307/AI-Weather-Wise-API
Programming Language(s)	Python 3.11
Framework(s) Used	FastAPI, Uvicorn, OpenAI SDK, Requests
Key Features Implemented	• Real-time weather data aggregation from OpenWeatherMap API • AI-powered predictive activity & clothing recommendations • Structured JSON API response endpoints for easy mobile/web integration • JWT token-based API authentication & rate-limiting • Redis response caching layer for optimized performance
Pending / Incomplete Features	None (All core phase functional requirements implemented)
Setup / Run Instructions	1. Clone repository: `git clone https://github.com/SWTID-2026-4307/AI-Weather-Wise-API.git` 2. Install dependencies: `pip install -r requirements.txt` 3. Configure environment variables in `.env` file (API Keys) 4. Run application: `uvicorn main:app --reload --port 8000`

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes

S.No	Criteria	Status (Yes / No)
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The AI Weather Wise API backend has been successfully unit tested and optimized for speed and reliability. Automated API documentation is accessible via Swagger UI at `http://localhost:8000/docs` after launching the server.